HABIB UNIVERSITY

Computer Architecture
EE 371 / CS 330 / CE 321
Spring 2024

> *"The microprocessor is a miracle." ~ Bill Gates*
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> *"You don't have conversations with microprocessors. You tell them what to do, then helplessly watch the disaster when they take you literally!" ~ David Brin*
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> *"Every time you turn on your new car, you're turning on 20 microprocessors. Every time you use an ATM, you're using a computer. Every time I use a settop box or game machine, I'm using a computer. The only computer you don't know how to work is your Microsoft computer, right?" ~ Scott McNealy*

Instructors:	Dr. Tariq Kamal (tariq.kamal@sse.habib.edu.pk) Office: W-224, Ext: 5266 Office Hours: TBA Section: L3
	Dr. Farhan Khan (farhan.khan@sse.habib.edu.pk) Office: W-222, Ext: 5226 Office Hours: MonTuesWed 10:00 – 11:00am Section: L1 and L6
	Dr. Waseem Hassan (waseem.hassan@sse.habib.edu.pk) Office: E-114 Ext: 5279 Office Hours: TBA Section: L2 and L4
Class Location/ Timings:	L1: W-321 every TuesThurs at 11:30am-12:45pm L2: W-321 every MonWed at 4:00pm-5:15pm L3: E-011 every MonWed at 8:30am-9:45am L4: H-509 every MonWed at 10:00am-11:15am L6: W-242 every MonTues at 8:30am-9:45am
Course Credit Hours:	3 hrs (with co-requisite +1 Lab credit hour separately)
Course Co-requisites:	EE371L/CE330L/CE321L Computer Architecture Lab
Course Prerequisites:	EE172/CS130 Digital Logic Design
Course LMS URL:	L1: https://hulms.instructure.com/courses/3521 L2: https://hulms.instructure.com/courses/3523 L3: https://hulms.instructure.com/courses/3525 L4: https://hulms.instructure.com/courses/3527 L6: https://hulms.instructure.com/courses/3531

Software Prerequisites:	Canvas, Zoom, MS teams, Acrobat Reader, MS office, Internet browser
Content Area:	This course meets requirements for CE/CS Major and EE Elective
Campus Safety Policy:	Please read the campus safety policy and protocols if the classes are in-person

I. Rationale:

It is important for the students to know about the hardware their programs run on. This under the hood knowledge is needed not only for those who want to design processors but also for those who want to write efficient code. The end of steady increase in clock speeds and the limitations of Moore's law would require innovative methods of processor design if we want increased performance from our processors.

II. Course Aims and Outcomes:

A. ***Aims***

The students will learn basic principles of computer architecture. How Moore's law has traditionally helped in performance increase and the effect of impending end. They'd also learn about RISC architectures, their assembly language and how to write programs in it. They'll also learn the design of a pipelined RISC processor. Caches, their working and their effect on program performance will also be discussed. This will be accomplished by integrating theory and practice through coordinated lecture and lab hours.

B. ***Program Learning Outcomes (PLO's)***

Program Learning Outcomes, are statements that describe what students will know and be able to do at the time of graduation. Based on the high-level objectives, the course facilitates and contributes to the following three Program Learning Outcomes:

1. PLO 1 – Engineering Knowledge
2. PLO 2 – Problem Analysis
3. PLO 3 – Design and Development of Solutions

C. ***Specific Student/Course Learning Outcomes (SLOs/CLOs):***

SLO/ CLO Description of SLO/CLO	Learning Domain Level	PLO
1. ***Explain*** the role of ISA in modern processors and instruction encodings and assembly language programming	Cog2	PLO1
2. ***Explain*** the architecture and working of a single cycle processor	Cog2	PLO1
3. ***Design*** the architecture to mitigate issues of a pipelined processor	Cog6	PLO3
4. ***Analyze the*** performance of cache operations	Cog4	PLO2

D. ***SLOs/CLOs to PLOs' mapping:***

	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1	50%	50%		
PLO 2				100%
PLO 3			100%	

III. Format and Procedures:

1. Medium of Instruction
   Every week two 75-minute sessions will be held in-person
2. Reference Books, Presentations and Support Material
   a. All course resources (presentations, recorded sessions, reference books, articles and all other support material) can be accessed through the course LMS site on Canvas.
   b. All assignments, quizzes and assessments with instructor solution would be managed at the course LMS site
3. Net-etiquettes and Participation Rules
   a. Students have to show active participation in class.
   b. In case we go online, students need to keep their microphone and camera ready to be used. Students will be required to keep their microphone and camera on during Q&A part of the lecture.
   c. Further Do's and Don'ts can be discussed in the first two weeks of classes.
4. Online Assessment Rules (if applicable)
   a. As per instructors directive, students may be required to turn on the camera during assessments.
   b. All working need to be submitted in required format, e.g. scanned PDF, with correct page sequence.
   c. Errors and omissions will not be entertained after assessment submission.

IV. Course Requirements:

Course readings:

(a) Required text:
   a. Computer Organization and Design, A Hardware/Software Interface. First Edition. David A. Patterson and John L. Hennessy

(b) Reference text:
   a. Computer Architecture A Quantitative Approach. Sixth Edition. John L. Hennessy and David A. Patterson

V. Assessments and Grading Procedures:

A. Assessments

1. Grade division:
   - Grades will be based on following assessments and their weights:

Theory	Assignments	20%
	Quizzes	20%
	Class Participation	10%
	Mid Term Exam	25%
	Final Exam	25%

2. Late Submission Policy:
   - For all assessments, 20% deduction per day late submission will be applicable or as conveyed by the course instructor.

3. Final Exam Policy
   - Please refer to the final exam policy.

B. Mapping of Assessments to SLOs/CLOs

	CLO1	CLO2	CLO3	CLO4
Assignment 1	X			
Assignment 2		X		
Assignment 3			X	
Assignment 4				X
Quiz 1	X			
Quiz 2		X		
Quiz 3			X	
Quiz 4				X
Midterm	X	X		
Final Exam			X	X
Class Participation				

** Mapping of Quiz/Assignment/Exam with CLOs may be revised*

C. Grading Scale:

Habib University Grading Scale GB4 will be applicable for this course (effective Fall2021)

Letter Grade	GPA Points	Percentage
A+	4.00	[95 – 100]
A	4.00	[90 – 95)
A-	3.67	[85 – 90)
B+	3.33	[80 – 85)
B	3.00	[75 – 80)
B-	2.67	[70 – 75)
C+	2.33	[67 – 70)
C	2.00	[63 – 67)
C-	1.67	[60 – 63)
F	0.00	[0, 60]

Note: $[a, b)$ is a range of numbers from a to b where a is included in the range and b is not.

VI. Attendance Policy:

Students are expected to attend all sessions. ***For all classes, university standard attendance policy of 85% is applicable. This means that the student can miss up to 4 classes at max, otherwise he/she will be automatically dropped from the course***. The instructor may also measure attendance in dynamic ways including in class participation, feedback on recorded sessions, performance in assessments etc. Students failing to join sessions must inform their instructor within 24 hours along with the reason. Student who does not meet minimum attendance threshold will be reported to OAP and dropped from the course.

VII. Accommodations for students with disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

VIII. Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

IX. Office hours:

During these hours the course instructor will be available to answer questions or provide additional help. Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.

X. Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work (unless group work is explicitly been allowed).

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

PLAGIARISM: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

CHEATING: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.

COLLUSION: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.
All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

XI. Week-wise Schedule (Tentative)

Timeline	Readings to be discussed	Assessment
Week - 1 January 08-12 (*First day of Classes: January 08, 2024*)	Welcome to the course. Course Syllabus, Policies and Overview	
Week – 2 January 15 – 19	Chapter 1 (*Book: P&H*) – Moore's law, Power Wall, Processor trends and Cost – Classes of Computer, RISC vs CISC 1.1, 1.2, 1.3, 1.4, 1.5	
Week - 3 January 22 – 26	Chapter 1 (*Book: P&H*) – Processor performance metrics (execution time, response time, CPI, etc.) 1.6, 1.7, 1.8 Chapter 2 (*Book: P&H*) – ISA, Need of Registers and Memory, Data width, instruction size, Memory and Register requirement for RISC-V 2.1, 2.2, 2.3	Assignment-1 (Out)
Week - 4 January 29 – February 2	Chapter 2 (*Book: P&H*) – Data Truncation and Sign Extension, Instruction Formats and Instruction Encoding – Logical and Conditional Instructions 2.4, 2.5, 2.6, 2.7	Assignment-1 (Due)
Week - 5 February 5 – 9	Chapter 2 (*Book: P&H*) – Procedures and Stack – Example codes of Procedures – Starting a program – Array vs pointers 2.8, 2.12, 2.14	Quiz-1
Week - 6 February 10 – 16	Chapter 3 (*Book: P&H*) – Arithmetic Unit for a Processor 3.1, 3.2, 3.3, Appendix A-5 Chapter 4 (*Book: P&H*) – Basic Units Required for a Processor 4.1, 4.2	Assignment-2 (Out)
Week - 7 February 19 – 23	Chapter 4 (*Book: P&H*) – Connecting Units to design a basic instruction processor – Designing a Data Path of a Single Cycle Processor 4.3, 4.4	
Week - 8 February 26 – March 2 (*Midterm Exam Week*)	Chapter 4 (*Book: P&H*) – Control Path of a Single Cycle Processor – Data path analysis for latency	Assignment-2 (Due) Quiz-2

Week – 9 March 4 – 8 (*Midterm Exam Week*)	Chapter 4 (*Book: P&H*) – Adding more instruction support in Single Cycle Processor – Pipelined RISC-V Processor Basics 4.4, 4.5	Midterm Exam (on deans' approval)
Week – 10 March 11 – 15	Chapter 4 (*Book: P&H*) – Hazards in Pipelined Processor (**1st Ramzan: March 11, 2024)* 4.7	Assignment-3 (Out) – in group of two
Week – 11 March 18 – 22	† Conference Days: March 22-24, 2024 † Pakistan Day: March 23, 2024 – Data Path of a Pipelined Processor 4.6	
Week – 12 March 25 – March 29	Chapter 4 (*Book: P&H*) – Data Path of a Pipelined Processor - Pipeline Architecture - Mitigation of Data Hazards – Control Path of a Pipelined Processor 4.6, 4.7	
Week - 13 April 1 – 5	Chapter 4 (*Book: P&H*) – Complete Pipeline Architecture with Mitigation of control Hazards 4.7, 4.8 Chapter 5 (*Book: P&H*) – Memory Hierarchy, Memory Types, Access Time 5.1, 5.2, 5.3	Assignment-3 (Due) Quiz-3 Assignment-4 (Out) – in group of two
Week - 14 April 8 – 12	Chapter 5 (*Book: P&H*) – Locality vs Spatiality, Direct Mapped Cache – Measuring Cache Performance Analysis (Hits vs Misses) † Eid ul Fitr: April 10 to April 12, 2024 5.4	
Week - 15 April 15 – 19	Chapter 5 (*Book: P&H*) – Associative Cache and Associativity – Replacement Policies, Cache Coherence 5.4	Assignment-4 (Due)
Week – 16 April 22 – 26	– Advanced Topics in Computer Architecture (Parallelism) 5.8	Quiz-4
	April 27 – 29, 2024 (Reading Days) April 30 & May 2 – 3 & 6 – 8, 2024	Final Term Exam